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# =====
# Lab 2: Retail Store Sales Analysis
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# 1. Create Sales Data
products <- c("Rice", "Beans", "Oil", "Soap")
sales    <- c(120, 80, 150, 60)

# Matrix (rows = products, cols = days)
sales_matrix <- matrix(c(120,130,125,
                        80, 90, 85,
                        150,160,155,
                        60, 70, 65),
                      nrow = 4, byrow = TRUE)

rownames(sales_matrix) <- products
colnames(sales_matrix) <- c("Mon", "Tue", "Wed")
print(sales_matrix)

# =====
# 2. List Structure
# =====
store_data <- list(products = products,
                   weekly_sales = sales_matrix)
print(store_data)

# =====
# 3. Conditional Filtering (High Sales)
# =====
high_sales <- c()

for (i in 1:length(sales)) {
  if (sales[i] > 100) {
    high_sales <- c(high_sales, products[i])
  }
}

print(high_sales)

# =====
# 4. Convert to Data Frame
# =====
sales_df <- data.frame(products, sales)
print(sales_df)

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# =====  
# 5. Visualization  
# =====  
  
# Line plot (trend)  
plot(sales, type = "o", col = "blue",  
      main = "Sales Trend", xlab = "Products", ylab = "Sales")  
  
# Pie chart  
pie(sales, labels = products, col = rainbow(length(products)))
```